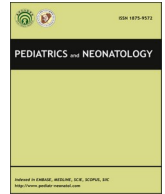




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Review Article

Impact of caffeine on neurodevelopmental outcomes in preterm infants: A systematic review and meta-analysis

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ABSTRACT

Although caffeine is among the commonly used medications in preterm infants, its long-term neurodevelopmental effects have remained uncertain. This systematic review and meta-analysis examined the potential long-term impacts of caffeine administration on the neurodevelopmental outcomes of preterm infants. A comprehensive search was conducted across PubMed, EMBASE, Google Scholar, Web of Science (ISI), Scopus, and the Cochrane Library. Eligible studies included randomized clinical trials or longitudinal studies in which the effects of caffeine administration on neurodevelopmental outcomes were compared to those of placebo or a control group. Neurodevelopmental outcomes, specifically motor, language, cognition, and cerebral palsy, were extracted from the eligible studies and pooled using STATA16 with a random-effects model. Egger's regression test was applied to ascertain the potential for publication bias. Among nine studies comprising 7,041 assessments, caffeine indicated a significant positive effect on motor development (WMD = 0.68, 95% CI: 0.27 to 1.09, $P < 0.001$, $I^2 = 0.00\%$) and cognitive outcomes (WMD = 2.85, 95% CI: 1.63 to 4.08, $P < 0.001$, $I^2 = 0.00\%$). Nevertheless, no significant impact was found on language development (WMD = 0.95, 95% CI: 0.37 to 2.27, $P = 0.16$, $I^2 = 0.00\%$). Further, the odds of cerebral palsy diminished by 35% by the use of caffeine (OR = 0.65, 95% CI: 0.53 to 0.79, $P < 0.001$, $I^2 = 0.00\%$). No publication bias was detected in any of the quantitative analyses. Both standard and high caffeine doses improved cognitive development but had no influence on language development.

In conclusion, caffeine may have beneficial effects on neurodevelopment, particularly motor and cognitive development and may also lower the risk of cerebral palsy.

1. Introduction

Preterm birth remains a global health concern, imposing a significant burden of disease on both families and healthcare systems [1,2]. In 2020, the global prevalence of preterm birth was estimated at 9.9%, accounting for nearly 13.4 million live births before 37 weeks of gestation, with estimates ranging from 12.3 to 15.2 million. Among these, nearly 15%, approximately 2 million infants, were born before 32 weeks of gestation, a critical threshold linked to a significantly elevated risk of mortality as well as severe long-term complications and

disabilities [3,4]. While advancements in neonatal care have enhanced survival rates, preterm infants remain at high risk for a range of health conditions, highlighting the constant need for enhanced interventions [5].

A major long-term complications of preterm birth is impaired neurodevelopment, which can manifest as delays in cognitive function, motor impairments, language deficits, and behavioral disorders [6–8]. Given the profound impact of preterm birth on neurodevelopmental outcomes, there is a critical need for effective strategies to alleviate these disabilities through both antenatal and postnatal interventions

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[5]. Antenatal interventions, such as the administration of steroids and magnesium sulfate, are essential components of prenatal medical care to protect fetal brain development and lowering the risk of complications associated with preterm birth. Postnatally, strategies such as family-centered care, skin-to-skin contact, kangaroo care, and optimized nutrition play a key role in fostering healthy neurodevelopment. These comprehensive approaches are fundamental in ameliorating long-term neurological and developmental outcomes for preterm infants [5,9].

Caffeine is widely recognized as a valuable medication in the neonatal care, especially for its role in managing apnea of prematurity as well as facilitating extubation in preterm infants [10,11]. Caffeine exerts its effect through stimulating the medullary respiratory center, enhancing sensitivity to carbon dioxide molecules, boosting the tone of skeletal muscles involved in the respiration, and ameliorating diaphragmatic contractility. Together, these mechanisms contribute to an overall increase in respiratory drive and oxygen consumption, which are essential for the respiratory stability of preterm infants [12]. While caffeine has been well established for its respiratory advantages in neonatal care, its long-term neurodevelopmental effects remain an area of ongoing research. A growing body of evidence suggests that caffeine may have beneficial effects on neurodevelopment in preterm infants, through uncertainties still exist [13–16]. In light of these considerations, this systematic review and meta-analysis aimed to provide a comprehensive assessment of caffeine's potential neuroprotective role, elucidating its impact on long-term outcomes and offering hope for significantly improved prognoses in preterm infants.

2. Methods

We undertook this review study following the Cochrane Collaboration's recommendations and reported it in accordance with the guidelines outlined by the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) checklist [17,18] (Supplementary file). The protocol for this meta-analysis has been registered with PROSPERO under the registration number CRD42025636675.

2.1. Search strategy

A systematic literature search was performed across PubMed, EMBASE, Google Scholar, Web of Science (ISI), Scopus, and the Cochrane Library using a combination of keywords related to caffeine (caffeine, caffeine citrate, and xanthine), child development (developmental disorder, language development, and cognitive development), and premature infants (preterm infant, neonatal prematurity, newborn, and extremely premature infant). The search was undertaken on January 30, 2025 (Supplementary File). Further, the reference lists of the relevant articles, the final included studies as well as secondary relevant studies, were reviewed to ensure no eligible studies were overlooked.

2.2. Eligibility criteria

We included randomized controlled trials (RCTs) along with longitudinal studies that had examined the effects of caffeine administration compared to a placebo or no-caffeine group on neurodevelopmental outcomes or explored the impact of different caffeine doses (standard vs. high) on neurodevelopment. High-dose caffeine generally refers to a loading dose of 80 mg/kg, followed by a maintenance dose of 20 mg/kg/day. Meanwhile standard (low) dose caffeine referred to a loading dose of 20 mg/kg, followed by a maintenance dose of 5–10 mg/kg/day. Further, early initiation is defined as starting caffeine within 2 days of life, whereas late initiation is defined as beginning after 2 days of life.

Cerebral palsy is currently defined as a group of lasting — though not necessarily static — disorders affecting movement, posture, and motor abilities. These conditions arise from a non-progressive disruption, injury, or abnormal development of the immature brain [19,20].

Eligible studies had focused on premature infants as the study population. The primary neurodevelopmental outcomes appraised were motor, cognitive, and language development, whereas secondary outcomes included the incidence of cerebral palsy. Studies were excluded if they were single-arm clinical trials or double-arm clinical trials in which the comparison group had received an intervention other than a placebo, no caffeine, or a specified caffeine dose (standard or high), as well as cross-sectional studies. We also excluded letters to the editor, commentaries, correspondences, conference abstracts, case reports, case series, and studies from which the required data or results could not be extracted.

2.3. Study screening and selection

Two independent reviewers initially screened the titles, abstracts, and keywords of records in order to identify potentially eligible studies. For articles where eligibility could not be specified based on the title or abstract alone, a full-text review was performed. Any disagreements between reviewers were resolved through discussion between these two independent reviewers or, if necessary, through consultation with the corresponding author.

2.4. Data extraction

Key information was extracted from the included studies as follows: the publication year, country, study design and methodology, sample size, study population characteristics, intervention groups (caffeine vs. placebo or no caffeine), duration and timing of follow-up, assessment tools for neurodevelopmental outcomes, ascertained outcomes (motor, cognitive, language development, and cerebral palsy incidence), and data required to estimate effect sizes. For ensuring consistency and accuracy, two reviewers independently performed the data extraction.

2.5. Assessment tools

There were different questionnaires: The BSID-III is a 30–90-min test appraising neurodevelopment in children up to 42 months, especially preterm or high-risk infants. It ascertains cognition, communication (receptive/expressive), and motor skills (fine/gross) [21]. The Gross Motor Function Classification System (GMFCS) was employed for motor impairment evaluation [22]. Wechsler Preschool and Primary Scale of Intelligence (WPPSI) measures verbal, performance (non-verbal), and full-scale IQ(23). It is to identify intellectual giftedness, delays, or learning disabilities. The Movement Assessment Battery for Children (MABC) is utilized to determine motor competence [23]. Clinical Evaluation of Language Fundamentals-Preschool (CELF-P) detects language disorders, delays, or auditory processing issues [24]. Finally, General Quotient in Griffiths Mental Development Scales(GQ) summarizes overall developmental progress [25].

2.6. Quality assessment

We appraised the quality of the included studies via two assessment tools: the Newcastle-Ottawa Scale (NOS) for observational studies and the Cochrane Risk of Bias Tool for RCTs [26,27]. With the NOS, the studies were ascertained based on three key domains: i) selection of the study groups, ii) comparability of these groups, and iii) ascertainment of exposure or outcome, with higher scores representing better methodological quality.

For RCTs, the Cochrane Risk of Bias Tool was employed for assessing methodological rigor across seven domains: generation of the random sequence, allocation concealment, participant and personnel blinding, outcome assessment blinding, completeness of outcome data, selective data reporting, and other potential biases in the study. Each of these domains was rated as having a low, high, or unclear risk of bias, thus

providing a comprehensive assessment of study reliability.

2.7. Data analyses

All statistical analyses were undertaken using STATA (Stata Corp., College Station, TX) version 16.0. For continuous outcome measures, we calculated the mean difference and standard deviation (SD) of change, while odds ratios (ORs) were derived for count variables from each study. Pooled effect sizes, including weighted mean differences (WMDs) and pooled ORs, were estimated via the random-effects model based on the DerSimonian–Laird method. For ascertaining the robustness of our findings, a series of sensitivity analyses was conducted using the leave-one-out method by which sequentially excluded each study to appraise its influence on the pooled effect sizes. Heterogeneity among the included studies was ascertained using Cochran's Q test and the I^2 statistic. Also, subgroup analyses were conducted based on moderating variables such as country, dosage (standard vs. high), and on whether they applied randomization. Egger's regression test was utilized to assess

the possibility of publication bias. A P-value of less than 0.05 was considered statistically significant.

3. Results

The comprehensive search performed across the databases initially identified 3729 articles. After removing duplicates and screening for eligibility, studies that did not fulfill our inclusion criteria were removed from the library. Ultimately, 9 articles [14,22–25,28–31] were selected for inclusion in the quantitative analyses (Fig. 1).

3.1. Characteristics of the eligible studies

Four of the selected studies were multicenter investigations [22,23,25,32], while the remaining studies have been conducted in specific countries: two in the USA [24,31], one in Turkey [28], one in Australia [29], and one in Canada [14]. The sample sizes varied significantly, ranging from 28 participants in a small clinical trial [28] to 2128

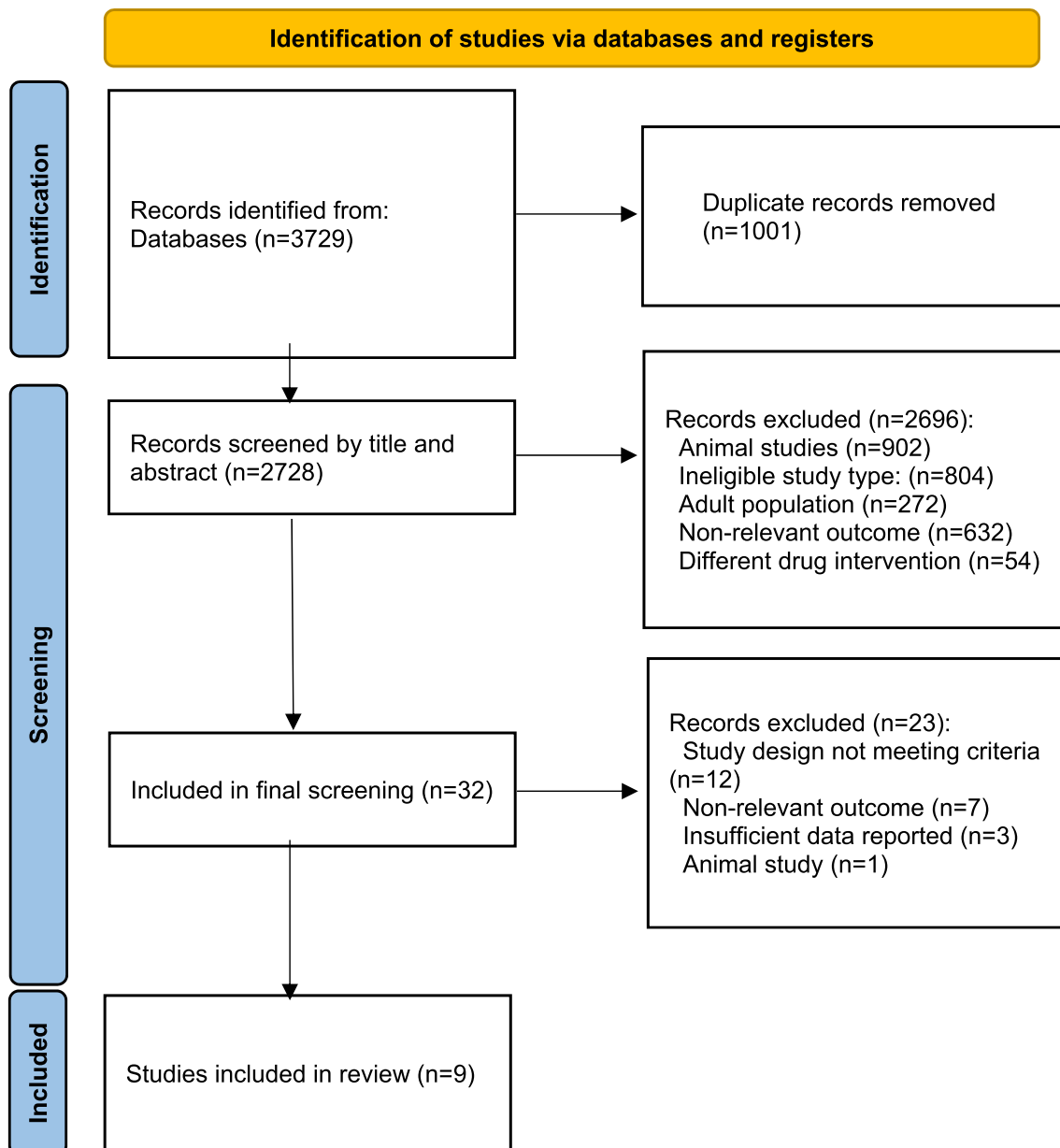


Fig. 1. Flow diagram depicting the study identification and selection process.

participants in a large cohort study [14], with a total of 7041 evaluations conducted across all studies. The duration of follow-up period varied among the included studies, spanning from 6 months to 11 years. Table 1 provides a detailed overview of the characteristics of these studies are provided in Table 1.

3.2. Meta-analysis results

3.2.1. Motor development

Using a random-effect model, our meta-analysis indicated a significant positive effect of caffeine on motor development in preterm infants (WMD = 0.68, 95% CI: 0.27 to 1.09, $P < 0.001$, $I^2 = 0.00\%$, across six studies, Fig. 2a). We also analyzed the impact of high-dose caffeine across three studies, which did not present any statistically significant differences (WMD = 0.38, 95% CI: 1.33 to 2.08, $P = 0.66$, Table 2).

3.2.2. Cognition

Our meta-analysis also exhibited a positive influence of caffeine administration in preterm infants on long-term cognitive development (WMD = 2.85, 95% CI: 1.63 to 4.08, $P < 0.001$, $I^2 = 0.00\%$, across six studies, Fig. 2b). Further subgroup analysis revealed that both standard-dose and high-dose caffeine had a significant positive impact on cognitive development. In studies ascertaining standard-dose caffeine (WMD = 2.75, 95% CI: 1.38 to 4.13, $P < 0.001$, two studies), the influence was statistically significant. Likewise, studies assessing high-dose caffeine also showed a significant positive effect (WMD = 3.17, 95% CI: 0.14 to 6.20, $P = 0.04$, four studies, Table 2).

3.2.3. Language

For this developmental domain, caffeine administration did not present a significant positive effect (WMD = 0.95, 95% CI: 0.37 to 2.27, $P = 0.16$, $I^2 = 0.00\%$, across five studies, Fig. 2c). Subgroup analysis by dosage showed no significant effect for either high-dose caffeine (WMD = 2.62, 95% CI: 1.73 to 6.97, $P = 0.24$, three studies) or standard-dose

caffeine (WMD = 0.78, 95% CI: 0.61 to 2.17, $P = 0.27$, two studies, Table 2).

3.2.4. Cerebral palsy

According to our findings, those who received caffeine along the neonatal period had 35% lower odds of developing CP when compared to the control group (OR = 0.65, 95% CI: 0.53 to 0.79, $P < 0.001$, $I^2 = 0.00\%$, Fig. 2d). Moreover, a significant protective impact for caffeine administration at standard-dose (OR = 0.65, 95% CI: 0.51 to 0.82, $P < 0.001$) was noted in the subgroup analysis. Only one study had specifically examined high-dose caffeine, and the effect was not statistically significant (OR = 0.73, 95% CI: 0.43 to 1.25, $P = 0.248$).

3.2.5. Sensitivity analysis

We ascertained the impact of each individual study on the pooled WMDs and ORs by systematically excluding each study from the meta-analysis. The sensitivity analysis indicated no substantial changes in the pooled effect sizes for motor, cognitive, and language development, as well as CP, reflecting the robustness of our findings (Fig. 3).

3.2.6. Publication bias

No evidence of potential publication bias was found across any of the evaluated domains, as determined by the Egger test. The corresponding p-values were 0.48 for motor development, 0.85 for cognitive development, 0.61 for language development, and 0.93 for CP. Funnel plots demonstrating these results are presented in Supplementary Fig. 1.

4. Discussion

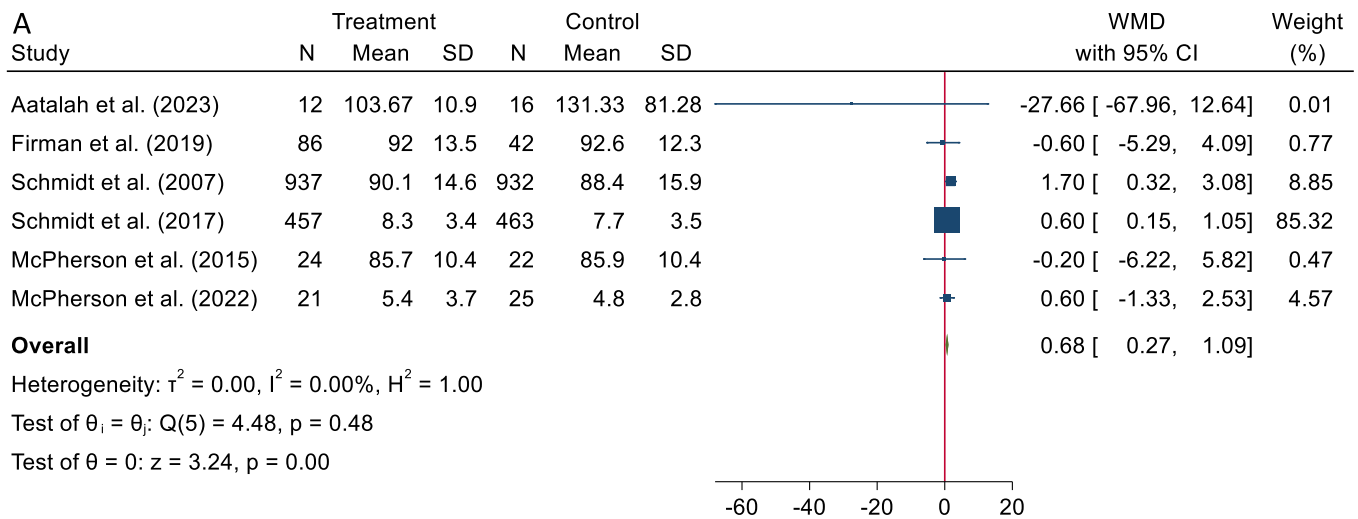
Recognizing the significance of long-term outcomes and neurodevelopment in preterm infants, and since caffeine is a widely used medication in this population, we designed this meta-analysis to appraise its effects on the neurodevelopment of preterm infants. To the best of our knowledge, this study was the first meta-analysis dedicated

Table 1

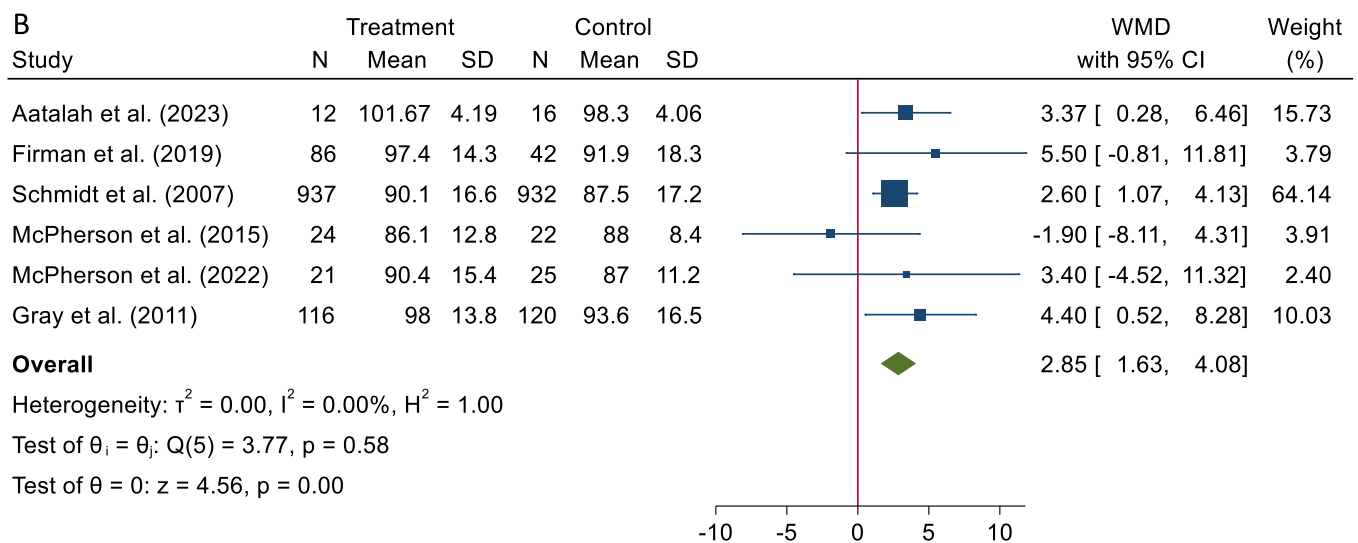
Characteristics of the studies included in the meta-analysis assessing the effect of caffeine on neurodevelopmental outcomes in preterm infants.

Authors	Publication year	Country	Total sample size (intervention, control)	Study design	Gestational age(week)	Birth weight case (gr) (case, control)	Age of participants	Questionnaire	Outcomes	Quality assessment
Aatalah et al. [28]	2023	Turkey	28 (12,16)	Clinical trial	28-34	1006 ± 266, 566 ± 487.6	6 months	BSID3	Motor Language Cognition	Some concerns
Firman et al. [29]	2019	Australia	128 (86,42)	Cohort	<28	852 ± 182, 934 ± 231	24 months	BSID3	Motor Language Cognition	High quality
Schmidt et al. [32]	2007	Multicenter	1869 (937,932)	RCT		961 ± 186, 952 ± 181	18-21 months	BSID3	Motor Cognition CP	Low
Schmidt et al. [22]	2012	Multicenter	1640 (833,807)	RCT		961 ± 186, 952 ± 181	5 years	GMFC, WPPSI	Language CP	Low to moderate
Schmidt et al. [23]	2017	Multicenter	920 (457,463)	RCT		961 ± 186, 952 ± 181	11 years	MABC2, GMFC	Motor CP	Moderate
Mcpherson et al. [31]	2015	USA	46 (24,22)	RCT	≤30	872 ± 257, 949 ± 245	24 months	BSID3	Motor Language Cognition	Low to some concerns
Mcpherson et al. [24]	2022	USA	46 (21,25)	RCT	≤30	817 ± 230, 995 ± 250	5 years	WPPSI, CELF-P2, MABC2	Motor Language Cognition	Moderate
Lodha et al. [14]	2018	Canada	2128 (1545,583)	Cohort	<29	950 ± 97.8, 860 ± 96.1	18-24 months	GMFC, BSID3	CP	High quality
Gray et al. [25]	2011	Multicenter	236 (116-120)	RCT	<30	1051 ± 281, 1032 ± 238	12 months	GQ	Cognition CP	Low to moderate concerns

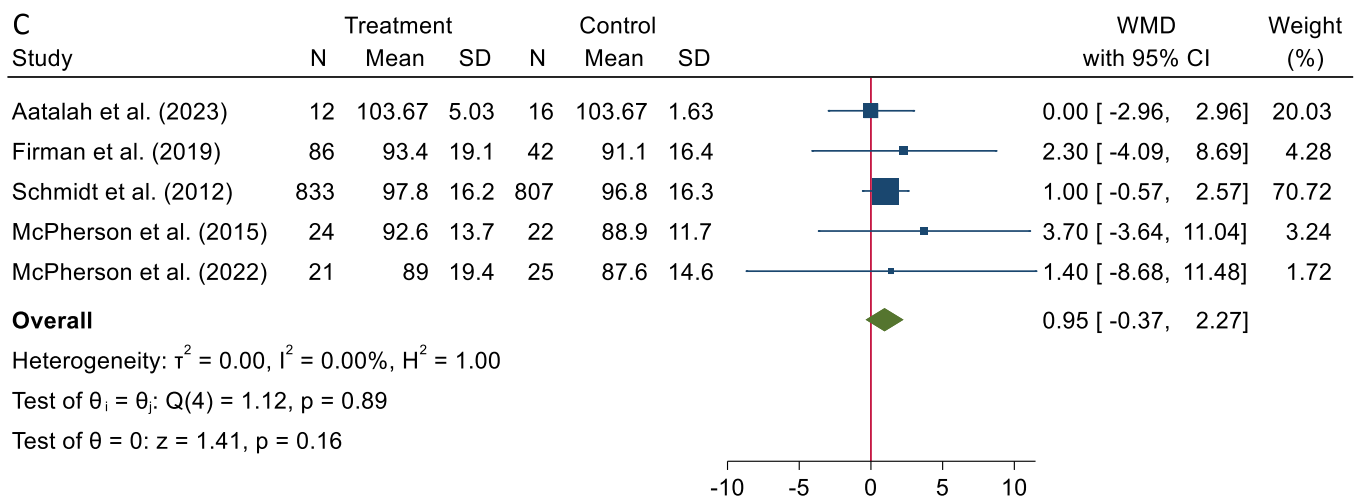
Abbreviation: BSID, Bayley Scales of Infant and Toddler Development; CELF-P2, Clinical Evaluation of Language Fundamentals Preschool-2; GA, gestational age, GMFC, Gross Motor Function Classification System; GQ, Graduation Questionnaire; MABC, Movement Assessment Battery for Children; RCT, Randomized Clinical Trial; WPPSI, Wechsler Preschool and Primary Scale of Intelligence; Wt, weight; USA, United States of America.



Random-effects DerSimonian–Laird model

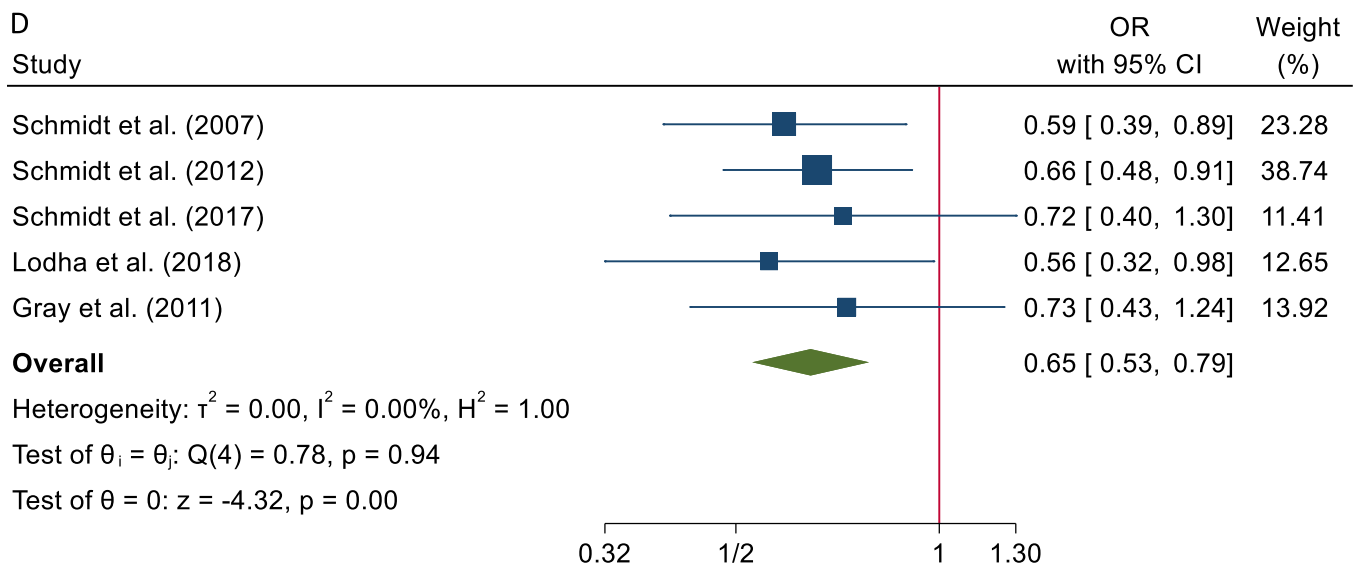


Random-effects DerSimonian–Laird model



Random-effects DerSimonian–Laird model

Fig. 2. Meta-analysis of the effect of caffeine on various neurodevelopmental outcomes in preterm infants. a: Motor development, b: Cognitive development, c: Language development, and d: Cerebral palsy. Each horizontal line represents an individual study, with squares indicating the effect size and diamonds representing the pooled effect estimate.



Random-effects DerSimonian–Laird model

Fig. 2. (continued).

Table 2

Subgroup analyses of the effects of caffeine on neurodevelopmental outcomes, including motor, cognitive, and language development in preterm infants.

Subgroups	Motor				Cognition				Language			
	Number of studies	WMD	95%CI	P-value	Number of studies	WMD	95%CI	P-value	Number of studies	WMD	95%CI	P-value
Country												
USA	2	0.53	-1.31, 2.36	0.57	2	0.16	-4.91, 5.22	0.95	2	2.90	-3.03, 8.84	0.33
Multicenter	2	0.95	-0.06, 1.95	0.06	2	2.84	1.42, 4.27	<0.001	1	1.00	-0.57, 2.57	0.21
Others	2	-6.42	-28.22, 15.37	0.56	2	3.78	1.01, 6.56	0.008	2	0.41	-2.28, 3.09	0.76
Implementation of randomization in the study												
Yes	4	0.69	0.28, 1.11	0.001	4	2.63	1.26, 3.99	<0.001	3	1.13	-0.39, 2.65	0.147
No	2	-6.42	-28.22, 15.37	0.56	2	3.78	1.01, 6.56	0.008	2	0.41	-2.28, 3.09	0.76
Caffeine dosage												
Standard dosage	3	0.98	-0.26, 2.21	0.12	2	2.75	1.38, 4.13	<0.001	2	0.78	-0.61, 2.17	0.27
High dosage	3	0.38	-1.33, 2.08	0.66	4	3.17	0.14, 6.20	0.04	3	2.62	-1.73, 6.97	0.24
Study design												
Trial	5	0.73	0.22, 1.25	0.005	5	2.75	1.50, 4.00	<0.001	4	0.89	-0.46, 2.24	0.19
Longitudinal	1	-0.60	-5.29, 4.09	0.80	1	5.50	-0.81, 11.81	0.08	1	2.30	-4.10, 8.69	0.48

specifically to this topic. Our findings confirmed the beneficial effect of caffeine on both motor and cognitive development, as well as a notable 35% decline in the incidence of CP among those who were given caffeine. Further, both standard and high doses of caffeine were associated with improved cognitive outcomes.

As our meta-analysis found, caffeine may have a significant positive influence on motor developmental outcomes. This impact is primarily mediated through its role as an adenosine receptor antagonist, especially within the neurotransmission pathways of the basal ganglia [33,34]. Through inhibiting A1 and A2A adenosine receptors, caffeine boosts dopaminergic activity, which is crucial for motor control [34]. This mechanism is responsible for its psychomotor stimulant properties, improving motor drive as well as overall performance [35,36]. Caffeine also influences muscle function and relaxation through optimizing the mobilization of calcium in the sarcoplasmic reticulum and regulating

sodium-potassium ATPase pump activity, resulting in improved muscle relaxation time [37,38]. It also plays a central role in energy metabolism by inhibiting cyclic adenosine monophosphate phosphodiesterase, particularly in the liver and skeletal muscle, thereby sustaining glycolysis and elevating blood glucose levels during exercise [38]. Further, caffeine may improve muscular strength by sensitizing calcium (Ca^{2+}) channels and promoting Ca^{2+} release from the sarcoplasmic reticulum [39,40]. Caffeine supplementation has also been found to promote Ca^{2+} release in a dose-dependent manner [39]. This mechanism is particularly relevant, as fatigue onset has been linked to reduced Ca^{2+} release and decreased myofibril Ca^{2+} sensitivity [41,42]. Through improving Ca^{2+} release, caffeine may help counteract these effects, thereby supporting better muscle function and endurance. In spite of these potential mechanisms underlying caffeine's influence on muscle function and motor development, challenges remain in determining the

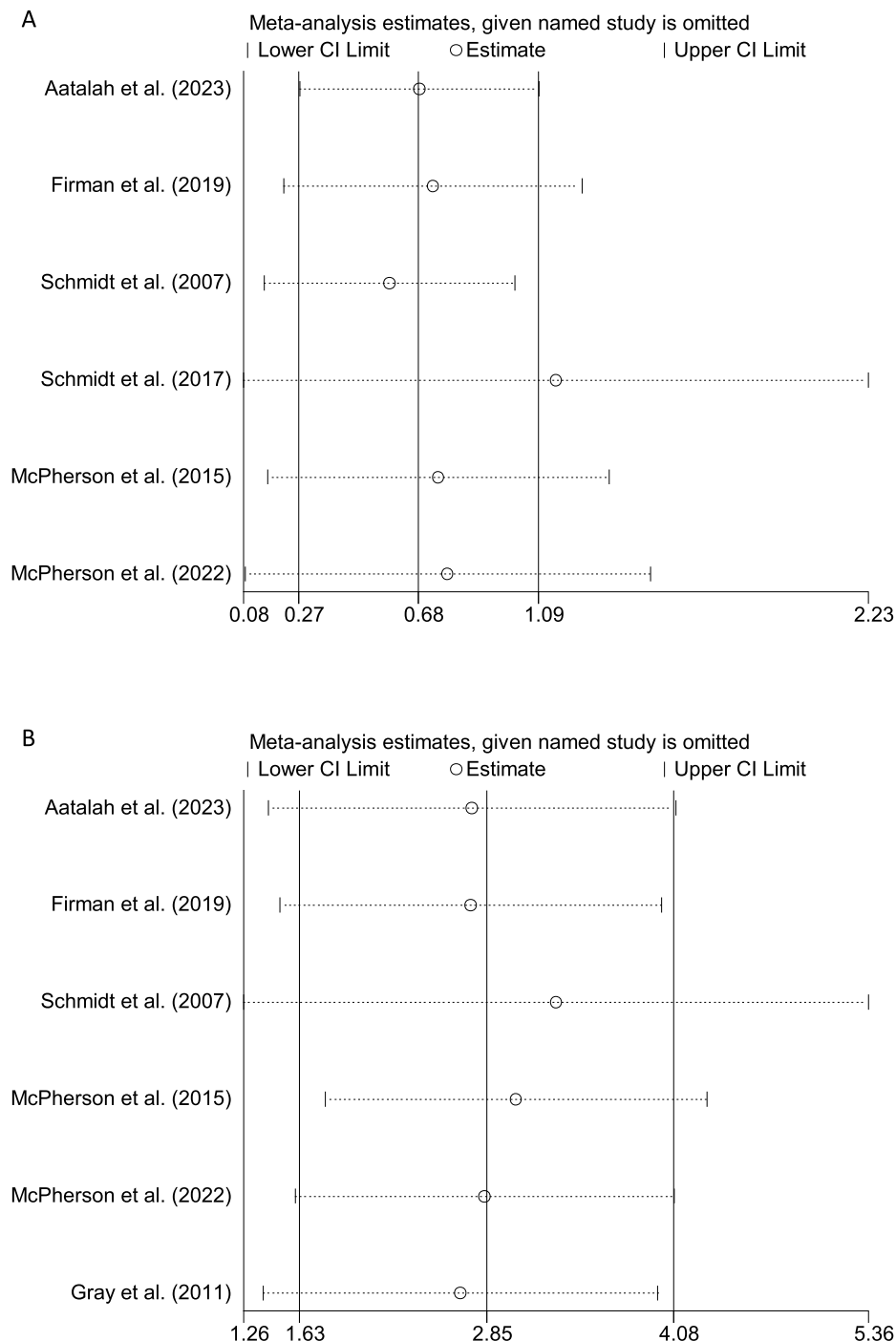


Fig. 3. Sensitivity analysis assessing the impact of individual studies on the overall effect size. a: Motor development, b: Cognitive development, c: Language development, and d: Cerebral palsy. The plot demonstrates that the exclusion of any single study does not significantly alter the pooled estimates.

optimal dosage and duration of caffeine consumption to maximize its benefits while minimizing potential risks [37,43,44].

Another significant benefit of caffeine is its long-term positive influence on cognitive outcomes. While research on this topic in infants remains limited, multiple mechanisms have been identified in adults. Caffeine boosts cognition primarily by antagonizing adenosine A1 receptors in the hippocampus and cortex, key regions in the brain involved in cognitive function [45,46]. This blockade prevents adenosine's inhibitory effects, thereby promoting neural excitability and augmenting information processing [47,48]. Caffeine's cognitive benefits may also be attributed to increased arousal, boosted behavioral efficiency,

and improved sensorimotor gating [48]. Evidence supporting the role of A1 receptor antagonism in cognitive enhancement includes findings from studies on ASP5854, a mixed A1/A2A receptor antagonist, which has been found to reverse the memory deficits induced by scopolamine in animal models [47]. While A2A receptor inhibition may also contribute to cognitive improvements, indicated by enhanced spatial memory in A2A receptor knockout mice, overexpression of these receptors has been linked to memory impairments [49,50].

Caffeine has also been reported to boost long-term memory and learning capacity [51], as well as slow the cognitive decline process [52]. Even though excessive caffeine intake may impair cognitive

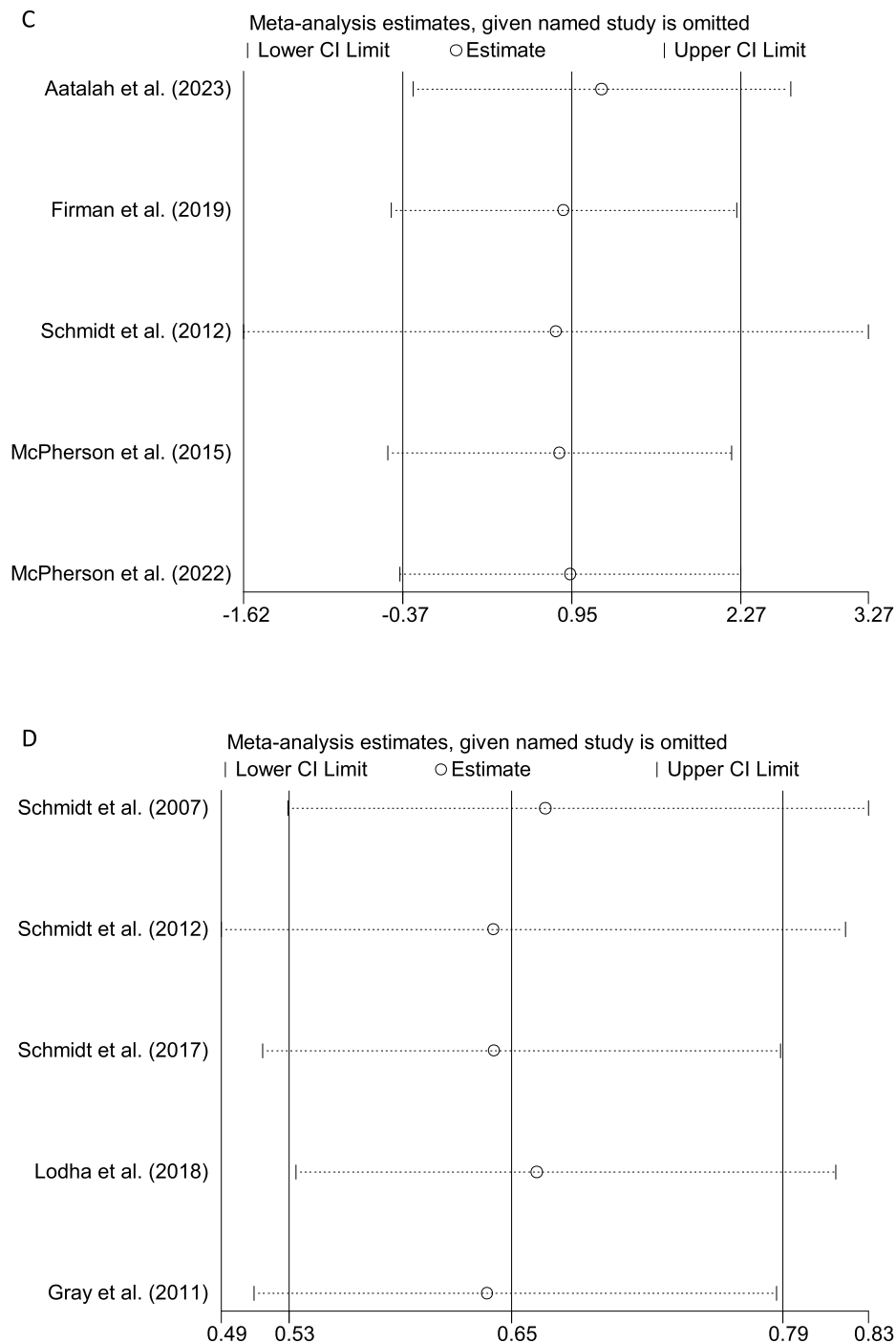


Fig. 3. (continued).

performance, especially in tasks requiring higher-order thinking, it also encourages synaptic plasticity through the release of brain-derived neurotrophic factor [53], which plays notable roles in learning and memory. Nevertheless, chronic caffeine use in adults has been linked to reduced gray matter volume, particularly in the medial temporal lobe, a region critical for memory and learning [54]. These findings highlight the complex association between caffeine consumption and cognitive function, emphasizing the need for further research, especially in pre-term neonates.

One of the key challenges in neonatology is determining the optimal caffeine dosage, standard versus high. A meta-analysis of six RCTs noted that high-dose caffeine administration was associated with lower rates of development of bronchopulmonary dysplasia and mortality but

showed no significant influence on neurodevelopmental outcomes [55]. Likewise, our analyses found that high-dose caffeine did not significantly influence motor and language development. However, for cognitive outcomes, both standard and high doses of caffeine were associated with ameliorated results.

CP is a heterogeneous condition caused by brain injury, primarily affecting movement, muscle tone, and motor skills [56,57]. It can have widespread consequences, affecting various aspects of life, including physical function (such as motor impairments, musculoskeletal deformities, and postural control) [58,59], as well as psychological and social well-being (with higher rates of anxiety, depression, and challenges related to employment and education) [60,61]. Our findings indicate lower odds of CP in preterm infants who received caffeine.

Since CP is a major healthcare concern often leading to lifelong disability and significant societal impact, preventive strategies aimed at lowering its incidence are particularly valuable.

In our study, caffeine therapy indicated a favorable impact on neurodevelopmental outcomes in preterm neonates, supporting its role as an important neuroprotective intervention. Beyond its well-established effectiveness in lowering the incidence of apnea of prematurity as well as facilitating earlier weaning from mechanical ventilation [62], caffeine has also been linked to improved neonatal morbidities. Previous studies have noted that caffeine use can lower the risk of bronchopulmonary dysplasia, and retinopathy of prematurity, while having no significant effect on intraventricular hemorrhage or necrotizing enterocolitis [63,64]. Even though some individual studies have reported potential gastrointestinal intolerance with caffeine [65], evidence from systematic reviews and meta-analyses suggests that caffeine is not significantly correlated with gastrointestinal complications [66]. Overall, adverse effects such as tachycardia [67] remain uncommon and manageable in clinical practice. Taken together, the therapeutic benefits of caffeine clearly outweigh its potential risks, highlighting its beneficial routine use in neonatal care.

Although the heterogeneity across all analyses was consistently low ($I^2 = 0\%$), this finding should be interpreted with caution given the evident clinical and methodological diversity among the included studies, including discrepancies in study design, follow-up duration, and neurodevelopmental assessment tools. One possible explanation for the observed low heterogeneity is the disproportionate impact of a limited number of large, high-quality studies, particularly the CAP trial series [22,23,30], which substantially contributed substantially to the pooled estimates. To address this concern, we undertook sensitivity analyses through sequentially excluding influential studies, including the CAP trials. The direction and magnitude of the pooled effect sizes remained largely unchanged, indicating that the overall results were not solely driven by any single study or study series. Nevertheless, the dominance of a limited number of influential trials may restrict the generalizability of the findings, especially to settings with different patient populations, intervention protocols, or follow-up strategies. Thus, in spite of the statistical homogeneity observed, the results should be interpreted in the context of the underlying clinical heterogeneity, and caution is warranted when extrapolating these findings to broader or heterogeneous populations.

As a key strength of this meta-analysis is that it was the first to specifically examine the effects of caffeine administration on neurodevelopment and occurrence of CP. Another notable advantage has been the consistency of findings, with minimal heterogeneity, no evidence of publication bias, and stable results across the included studies. Nonetheless, some limitations should be acknowledged. A major limitation of this meta-analysis is the substantial heterogeneity in both the timing of follow-up and the neurodevelopmental assessment tools employed across the included studies. Neurodevelopmental outcomes appraised at 6 or 12 months of corrected age are not directly comparable to those assessed at 18–24 months or at school age, as early evaluations have limited sensitivity for detecting higher-order cognitive, behavioral, and executive function impairments that may emerge later in childhood. Even though sensitivity analyses were performed to examine the robustness of our findings, they cannot fully overcome the inherent limitations of early neurodevelopmental assessments in representing long-term outcomes. Thus, the pooled effect sizes should be interpreted with caution, and the results should be perceived as reflecting early to mid-term neurodevelopmental trends rather than definitive long-term neurodevelopmental outcomes. In order to address these gaps, we recommend that future studies incorporate standardized, long-term follow-ups across multiple centers, applying a uniform assessment approach.

5. Conclusions

In conclusion, caffeine, a widely used medication in preterm infants, is linked to potentially favorable neurodevelopmental outcomes, particularly in motor and cognitive domains, as well as with a lower reported incidence of cerebral palsy in the included studies. Nevertheless, considering the heterogeneity in follow-up duration, assessment tools, and the predominance of a limited number of influential trials, these findings ought to be interpreted as associations rather than definitive causal effects. While caffeine remains an essential therapy for the management of apnea of prematurity, its possible long-term neurodevelopmental benefits warrant cautious consideration. The results highlight the need for additional large-scale, methodologically robust studies with standardized outcome measures and extended follow-up to elucidate the magnitude, durability, and causal nature of caffeine's potential neurodevelopmental effects in preterm infants.

Authors contribution

HF writing, editing and reviewing literature, RT conceptualization, analysis, writing and editing the manuscript. ShA conceptualization, writing, and editing. BI writing, editing and reviewing literature, KhSN writing, editing and reviewing literature. HB conceptualization, writing, editing and reviewing literature, supervision. The final version of the manuscript approved by all authors.

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Declaration of competing interest

Authors declare no conflict of interests.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.pedneo.2026.02.003>.

Data availability

All relevant data are within the paper and its Supporting Information files.

References

- [1] Walani SR. Global burden of preterm birth. *Int J Gynaecol Obstet* 2020;150:31–3.
- [2] Menon R. Preterm birth: a global burden on maternal and child health. *Pathog Glob Health* 2012;106:139–40.
- [3] Ohuma EO, Moller AB, Bradley E, Chakwera S, Hussain-Alkhateeb L, Lewin A, et al. National, regional, and global estimates of preterm birth in 2020, with trends from 2010: a systematic analysis. *Lancet* 2023;402:1261–71.
- [4] Behrman RE, Butler AS, editors. *Mortality and acute complications in preterm infants. Preterm birth: causes, consequences, and prevention*. Washington (DC): National Academies Press (US); 2007.
- [5] Molloy EJ, El-Dib M, Soul J, Juul S, Gunn AJ, Bender M, et al. Neuroprotective therapies in the NICU in preterm infants: present and future (Neonatal Neurocritical Care Series). *Pediatr Res* 2024;95:1224–36.
- [6] Hee Chung E, Chou J, Brown KA. Neurodevelopmental outcomes of preterm infants: a recent literature review. *Transl Pediatr* 2020;9(Suppl 1):S3–8.
- [7] Song IG. Neurodevelopmental outcomes of preterm infants. *Clin Exp Pediatr* 2022;66:281–7.
- [8] Kiechl-Kohlendorfer U, Ralsler E, Pupp Peglow U, Reiter G, Trawöger R. Adverse neurodevelopmental outcome in preterm infants: risk factor profiles for different gestational ages. *Acta Paediatr* 2009;98:792–6.
- [9] Boyd S, Tapawan SJ, Badawi N, Popat H. Protecting the brain of the micropreemie. *Semin Fetal Neonatal Med* 2022;27:101370.
- [10] Chavez L, Bancalari E. Caffeine: some of the evidence behind its use and abuse in the preterm infant. *Neonatology* 2022;119:428–32.
- [11] Armanian AM, Iranpour R, Faghiliani E, Salehimehr N. Caffeine administration to prevent apnea in very premature infants. *Pediatr Neonatol* 2016;57:408–12.

- [12] Abdel-Hady H, Nasef N, Shabaan AE, Nour I. Caffeine therapy in preterm infants. *World J Clin Pediatr* 2015;4:81–93.
- [13] Synnes A, Grunau RE. Neurodevelopmental outcomes after neonatal caffeine therapy. *Semin Fetal Neonatal Med* 2020;25:101160.
- [14] Lodha A, Entz R, Synnes A, Creighton D, Yusuf K, Lapointe A, et al. Early caffeine administration and neurodevelopmental outcomes in preterm infants. *Pediatrics* 2019;143:e20181348.
- [15] Zhang R, Manza P, Volkow ND. Prenatal caffeine exposure: association with neurodevelopmental outcomes in 9- to 11-year-old children. *J Child Psychol Psychiatry* 2022;63:563–78.
- [16] Chiu HY, Chu SM, Lin HY, Tsai ML, Chen YT, Lin HC. Evidence base multi-discipline critical strategies toward better tomorrow for very preterm infants. *Pediatr Neonatol* 2020;61:371–7.
- [17] Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ* 2021;372: n71.
- [18] Higgins JPT, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al. editors. *Cochrane handbook for systematic reviews of interventions*. vol. 4. Hoboken: Wiley; 2019.
- [19] Bax M, Goldstein M, Rosenbaum P, Leviton A, Paneth N, Dan B, et al. Executive committee for the definition of cerebral palsy. Proposed definition and classification of cerebral palsy. *Dev Med Child Neurol*. 2005;47:571–6.
- [20] Cans C, Dolk H, Platt MJ, Colver A, Prasauskiene A, Krägeloh-Mann I. Recommendations from the SCPE collaborative group for defining and classifying cerebral palsy. *Dev Med Child Neurol Suppl* 2007;109:35–8.
- [21] Del Rosario C, Slevin M, Molloy EJ, Quigley J, Nixon E. How to use the Bayley Scales of Infant and Toddler Development. *Arch Dis Child Educ Pract Ed* 2021;106: 108–12.
- [22] Schmidt B, Anderson PJ, Doyle LW, Dewey D, Grunau RE, Asztalos EV, et al. Survival without disability to age 5 years after neonatal caffeine therapy for apnea of prematurity. *JAMA* 2012;307:275–82.
- [23] Schmidt B, Roberts RS, Anderson PJ, Asztalos EV, Costantini L, Davis PG, et al. Academic performance, motor function, and behavior 11 years after neonatal caffeine citrate therapy for apnea of prematurity: an 11-Year Follow-up of the CAP randomized clinical trial. *JAMA Pediatr* 2017;171:564–72.
- [24] McPherson C, Lean RE, Cyr PEP, Inder TE, Rogers CE, Smyser CD. Five-year outcomes of premature infants randomized to high or standard loading dose caffeine. *J Perinatol* 2022;42:631–5.
- [25] Gray PH, Flenady VJ, Charles BG, Steer PA, Caffeine Collaborative Study Group. Caffeine citrate for very preterm infants: effects on development, temperament and behaviour. *J Paediatr Child Health* 2011;47:167–72.
- [26] Wells GA, Shea BJ, O'Connell D, Peterson J, Welch V, Losos M, et al. The Newcastle-Ottawa scale (NOS) for assessing the quality of nonrandomised studies in meta-analyses. In: 2. Ottawa Hospital Research Institute; 2011. p. 1–12.
- [27] Higgins JP, Altman DG, Gøtzsche PC, Jüni P, Moher D, Oxman AD. The cochrane collaboration's tool for assessing risk of bias in randomised trials. *BMJ* 2011;343: d5928.
- [28] Atalah YEY, Bariş HE, Akdere SK, Sabanci M, Özdemir H, Gücüyener K, et al. Effects of caffeine therapy for apnea of prematurity on sleep and neurodevelopment of preterm infants at 6 months of corrected age. *J Clin Sleep Med* 2023;19:2075–85.
- [29] Firman B, Molnar A, Gray PH. Early high-dose caffeine citrate for extremely preterm infants: neonatal and neurodevelopmental outcomes. *J Paediatr Child Health* 2019;55:1451–7.
- [30] Schmidt B, Roberts RS, Davis P, Doyle LW, Barrington KJ, Ohlsson A, et al. Long-term effects of caffeine therapy for apnea of prematurity. *N Engl J Med* 2007;357: 1893–902.
- [31] McPherson C, Neil JJ, Tjoeng TH, Pineda R, Inder TE. A pilot randomized trial of high-dose caffeine therapy in preterm infants. *Pediatr Res* 2015;78:198–204.
- [32] Schmidt B, Roberts RS, Davis P, Doyle LW, Barrington KJ, Ohlsson A, et al. Long-term effects of caffeine therapy for apnea of prematurity. *N Engl J Med* 2007;357: 1893–902.
- [33] Ferré S. Role of the central ascending neurotransmitter systems in the psychostimulant effects of caffeine. *J Alzheimers Dis* 2010;20(Suppl 1):S35–49.
- [34] Fisone G, Borgkvist A, Usiello A. Caffeine as a psychomotor stimulant: mechanism of action. *Cell Mol Life Sci* 2004;61:857–72.
- [35] Franco-Alvarenga PE, Brietzke C, Canestri R, Goethel MF, Viana BF, Pires FO. Caffeine increased muscle endurance performance despite reduced cortical activation and unchanged neuromuscular efficiency and corticomuscular coherence. *Nutrients* 2019;11:2471.
- [36] Bowtell JL, Mohr M, Fulford J, Jackman SR, Ermidis G, Krstrup P, et al. Improved exercise tolerance with caffeine is associated with modulation of both peripheral and central neural processes in human participants. *Front Nutr* 2018;5:6.
- [37] Ferreira LHB, Forbes SC, Barros MP, Smolarek AC, Enes A, Lancha-Junior AH, et al. High doses of caffeine increase muscle strength and calcium release in the plasma of recreationally trained men. *Nutrients* 2022;14:4921.
- [38] Graham TE, Battram DS, Dela F, El-Sohemy A, Thong FS. Does caffeine alter muscle carbohydrate and fat metabolism during exercise? *Appl Physiol Nutr Metab* 2008; 33:1311–8.
- [39] Klein M, Simon B, Schneider MF. Effects of caffeine on calcium release from the sarcoplasmic reticulum in frog skeletal muscle fibres. *J Physiol* 1990;425:599–626.
- [40] Daly J. Caffeine analogs: biomedical impact. *Cell Mol Life Sci* 2007;64:2153–69.
- [41] Allen DG, Lamb GD, Westerblad H. Impaired calcium release during fatigue. *J Appl Physiol* (1985) 2008;104:296–305.
- [42] Westerblad H, Lee JA, Lännergren J, Allen DG. Cellular mechanisms of fatigue in skeletal muscle. *Am J Physiol Cell Physiol* 1991;261:C195–209.
- [43] Astorino TA, Rohmann RL, Firth K. Effect of caffeine ingestion on one-repetition maximum muscular strength. *Eur J Appl Physiol* 2008;102:127–32.
- [44] Goldstein E, Jacobs PL, Whitehurst M, Penhollow T, Antonio J. Caffeine enhances upper body strength in resistance-trained women. *J Int Soc Sports Nutr* 2010;7:18.
- [45] Nehlig A. Is caffeine a cognitive enhancer? *J Alzheimers Dis* 2010;20(Suppl 1): S85–94.
- [46] Fredholm BB, Bättig K, Holmén J, Nehlig A, Zvartau EE. Actions of caffeine in the brain with special reference to factors that contribute to its widespread use. *Pharmacol Rev* 1999;51:83–133.
- [47] Mihara T, Mihara K, Yarimizu J, Mitani Y, Matsuda R, Yamamoto H, et al. Pharmacological characterization of a novel, potent adenosine A1 and A2A receptor dual antagonist, 5-[5-amino-3-(4-fluorophenyl) pyrazin-2-yl]-1-isopropylpyridine-2 (1H)-one (ASP5854), in models of Parkinson's disease and cognition. *J Pharmacol Exp Ther* 2007;323:708–19.
- [48] Ribeiro JA, Sebastiao AM. Caffeine and adenosine. *J Alzheimers Dis* 2010;20(Suppl 1):S3–15.
- [49] Giménez-Llort L, Schiffmann SN, Shmidt T, Canela L, Camón L, Wassholm M, et al. Working memory deficits in transgenic rats overexpressing human adenosine A2A receptors in the brain. *Neurobiol Learn Mem* 2007;87:42–56.
- [50] Wang JH, Ma YY, van den Buuse M. Improved spatial recognition memory in mice lacking adenosine A2A receptors. *Exp Neurol* 2006;199:438–45.
- [51] Fiani B, Zhu L, Musch BL, Briceno S, Andel R, Sadeq N, et al. The neurophysiology of caffeine as a central nervous system stimulant and the resultant effects on cognitive function. *Cureus* 2021;13:e15032.
- [52] Gardener SL, Rainey-Smith SR, Villemagne VL, Fripp J, Doré V, Bourgeat P, et al. Higher coffee consumption is associated with slower cognitive decline and less cerebral Aβ-Amyloid accumulation over 126 months: data from the Australian imaging, biomarkers, and lifestyle study. *Front Aging Neurosci* 2021;13:744872.
- [53] Lao-Peregrín C, Ballesteros JJ, Fernández M, Zamora-Moratalla A, Saavedra A, Gómez Lázaro M, et al. Caffeine-mediated BDNF release regulates long-term synaptic plasticity through activation of IRS2 signaling. *Addict Biol* 2017;22: 1706–18.
- [54] Lin YS, Weibel J, Landolt HP, Santini F, Meyer M, Borgwardt S, et al. Caffeine-induced plasticity of grey matter volume in healthy brains: a placebo-controlled multimodal within-subject study. *bioRxiv* 2019:804047.
- [55] Vliegthart R, Miedema M, Hutten GJ, van Kaam AH, Onland W. High versus standard dose caffeine for apnoea: a systematic review. *Arch Dis Child Fetal Neonatal Ed* 2018;103:F523–9.
- [56] Gulati S, Sondhi V. Cerebral palsy: an overview. *Indian J Pediatr* 2018;85:1006–16.
- [57] Aisen ML, Kerkovich D, Mast J, Mulroy S, Wren TA, Kay RM, et al. Cerebral palsy: clinical care and neurological rehabilitation. *Lancet Neurol* 2011;10:844–52.
- [58] Gaebler-Spira D, Green M. Cerebral palsy. *J Pediatr Rehabil Med* 2020;13:105–6.
- [59] Mezaal MA, Nouri KA, Abdool S, Al Safar K, Nadeem AS. Cerebral palsy in adults consequences of non progressive pathology. *Open Neurol J* 2009;3:24–6.
- [60] Bhatt N, Canella J, Gentile JP. Brief review: psychological health and life quality of cerebral palsy. *Theory and Clinical Practice in Pediatrics* 2022;4:102–5.
- [61] Lim MSY, Wong CP. Impact of cerebral palsy on the quality of life in patients and their families. *Neurol Asia* 2009;14:27–33.
- [62] Kumar VH, Lipshultz SE. Caffeine and clinical outcomes in premature neonates. *Children (Basel)* 2019;6:118.
- [63] Aranda JV, Beharry K, Valencia GB, Natarajan G, Davis J. Caffeine impact on neonatal morbidities. *J Matern Fetal Neonatal Med* 2010;23(Suppl 3):20–3.
- [64] Szatkowski L, Fateh S, Abramson J, Kwok TC, Sharkey D, Budge H, et al. Observational cohort study of use of caffeine in preterm infants and association between early caffeine use and neonatal outcomes. *Arch Dis Child Fetal Neonatal Ed* 2023;108:505–10.
- [65] Welsh C, Pan J, Belik J. Caffeine impairs gastrointestinal function in newborn rats. *Pediatr Res* 2015;78:24–8.
- [66] Gama B, von Hafe M, Vieira R, Soares H, Azevedo I, Rocha G. Gastrointestinal effects of caffeine in preterm infants: a systematic review and Bayesian meta-analysis. *Arch Dis Child Fetal Neonatal Ed* 2024;109:670–6.
- [67] Lenasi H, Rihar E, Filipić J, Klemenc M, Fister P. The effect of caffeine on heart rate variability in newborns: a pilot study. *Life (Basel)*. 2023;13:1459.